

OBJECT-ORIENTED MODELLING

Object-Oriented Programming

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2025-III



UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS

Outline

- 1 Creating Models in Design ✓
- 2 Evolution of Programming Languages ✓
- 3 Four Design Principles
- 4 SOLID Principles



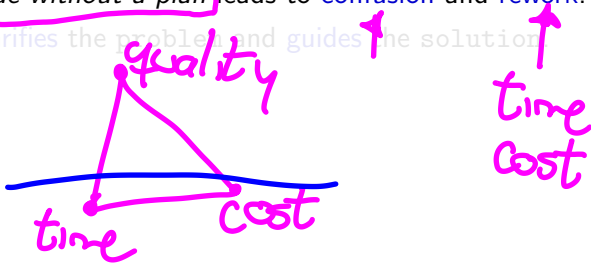
Outline

- 1 Creating Models in Design
- 2 Evolution of Programming Languages
- 3 Four Design Principles
- 4 SOLID Principles



Design Before Code

- **Design** should come before coding.
- Jumping into *code without a plan* leads to confusion and rework.
- Good design clarifies the problem and guides the solution.



Design Before Code

- **Design** should come **before coding**.
- Jumping into *code without a plan* leads to **confusion** and **rework**.
- **Good design** clarifies the problem and guides the solution.

1

2



Understanding the Requirements

- **Requirements** must be well understood before design.
- Ask questions, clarify ambiguities, and document all requirements.
- Requirements define the scope and direction of the design.

↑
list ⇒ need



Understanding the Requirements

- **Requirements** must be well understood before design.
- Ask questions, clarify ambiguities, and document all requirements.
- **Requirements** define the **scope** and **direction** of the design.

business vocabulary



Design Based on the Problem

- **Design** should be driven by the problem, not by technology.
- Focus on what *needs* to be solved, *not just* how to implement it.
- Use the **problem statement** to identify *key objects* and their *relationships*.



Design Based on the Problem

- **Design** should be driven by the problem, not by technology.
- Focus on what **needs** to be solved, *not just* how to implement it.
- Use the **problem statement** to identify **key objects** and their relationships.

non-technical

system



Object-Oriented Approach

- The **object-oriented approach** models the **system** as a **collection** of interacting **objects**.
- Each **object** represents a **real-world** entity or concept.
- **Objects** encapsulate **data** and **behavior**.

objects ← *elements*
relations



Conceptual Design and Technical Design

- **Conceptual Design**: What the **system should do**, using high-level models.

- **Technical Design**: How the system will be implemented, using detailed diagrams and specifications.

- Both are essential for a successful software project.

Diagramas
Referencia Document

UML
Unified modeling language
draw.io



Conceptual Design and Technical Design

- **Conceptual Design:** What the **system should do**, using high-level models.
- **Technical Design:** How the **system will be implemented**, using detailed diagrams and specifications.
- Both are **essential** for a **successful** software project.



Conceptual Design and Technical Design

- **Conceptual Design:** What the system should do, using high-level models.
- **Technical Design:** How the system will be implemented, using detailed diagrams and specifications.
- Both are essential for a successful software project.

USER

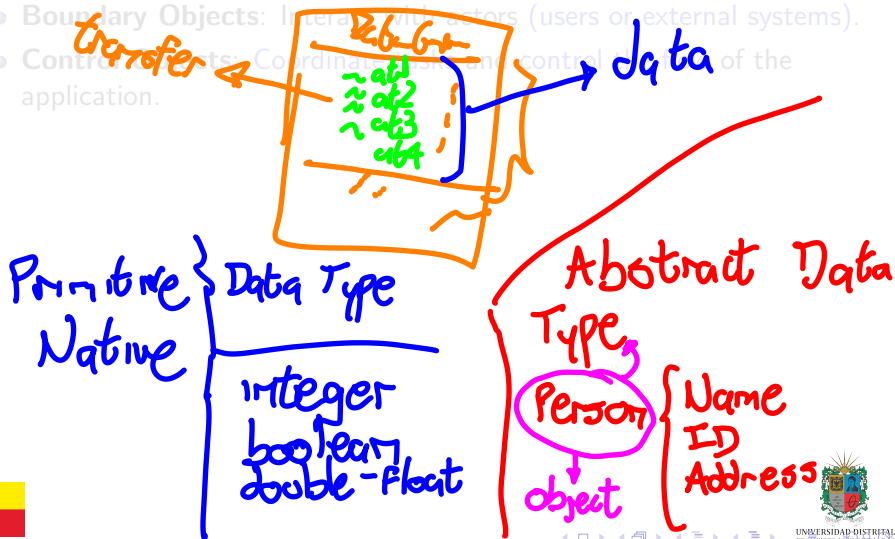
tech team

expectations



Categories of Objects

- **Entity Objects:** Represent information and data.
- **Boundary Objects:** Interact with actors (users or external systems).
- **Control Objects:** Coordinate tasks and control objects of the application.



Categories of Objects

- **Entity Objects:** Represent **information** and **data**.
- **Boundary Objects:** Interact with actors (**users or external systems**).
- **Control Objects:** Coordinate tasks and control the flow of the application.



Categories of Objects

- **Entity Objects:** Represent information and data.
- **Boundary Objects:** Interact with actors (users or external systems).
- **Control Objects:** Coordinate tasks and control the flow of the application.

logic → most of the objects



Documentation in Software

- **Documentation** is essential for communication and maintenance.
- Includes requirements, design diagrams, user manuals, and code comments.
- Good **documentation** helps new team members *understand* the *system* quickly.



Documentation in Software

- **Documentation** is essential for **communication** and **maintenance**.
- Includes requirements, design diagrams, user manuals, and code comments.
- Good **documentation** helps **new** team members *understand* the *system* quickly.



Documentation in Software

- **Documentation** is essential for communication and maintenance.
- Includes requirements, design diagrams, user manuals, and code comments.
- Good **documentation** helps new team members understand the system quickly.

old



solve problems

find a solution
innovative



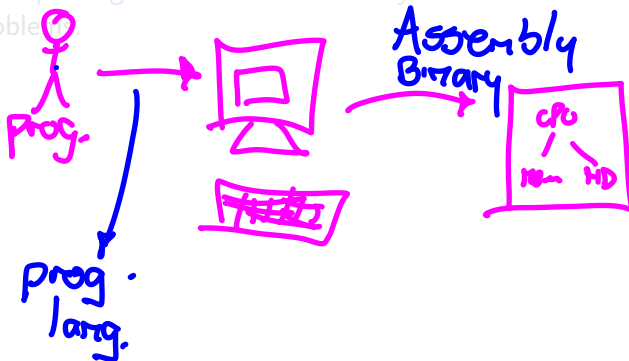
Outline

- 1 Creating Models in Design
- 2 Evolution of Programming Languages
- 3 Four Design Principles
- 4 SOLID Principles



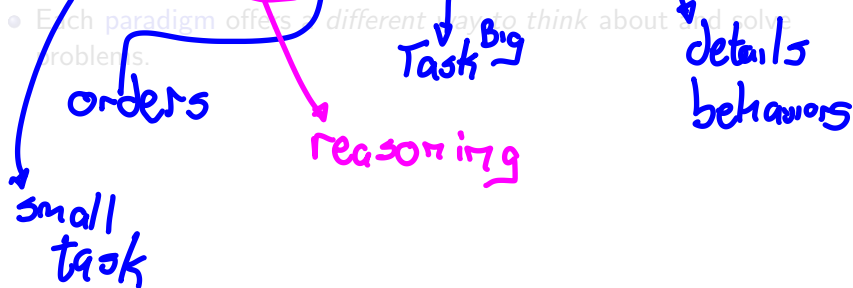
Talk with Machines: Programming Paradigms

- **Programming languages** are tools to **communicate with machines**.
- **Paradigms**: Imperative, Procedural, Object-Oriented, Functional, Logic
- Each paradigm offers a *different way to think about and solve problems*.



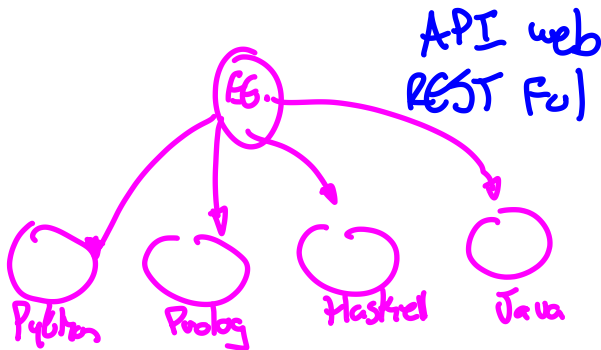
Talk with Machines: Programming Paradigms

- **Programming languages** are tools to communicate with machines.
- **Paradigms:** Imperative, Procedural, Object-Oriented, Functional, Logic.

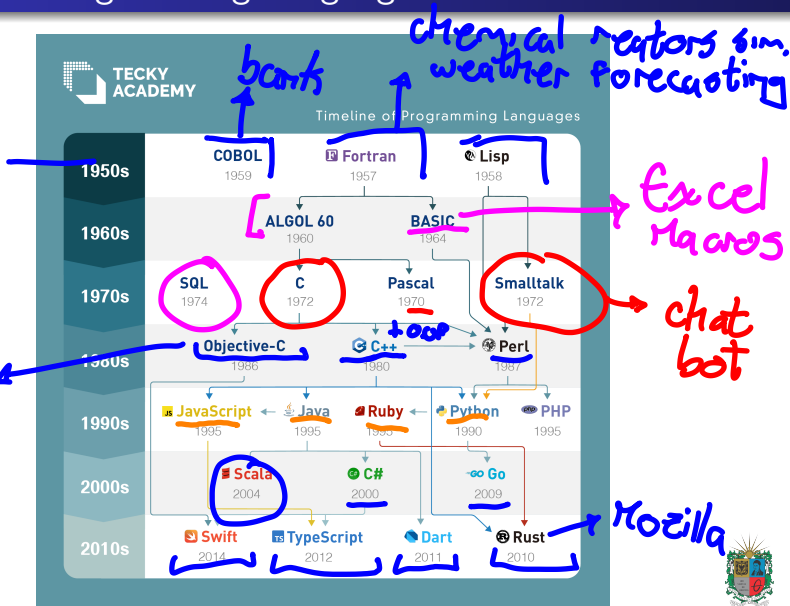


Talk with Machines: Programming Paradigms

- **Programming languages** are tools to **communicate with machines**.
- **Paradigms**: Imperative, Procedural, Object-Oriented, Functional Logic.
- Each **paradigm** offers a *different way to think* about and solve problems.

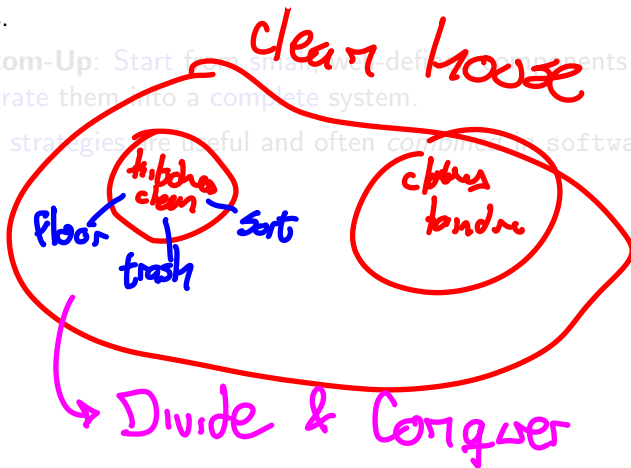


History of Programming Languages



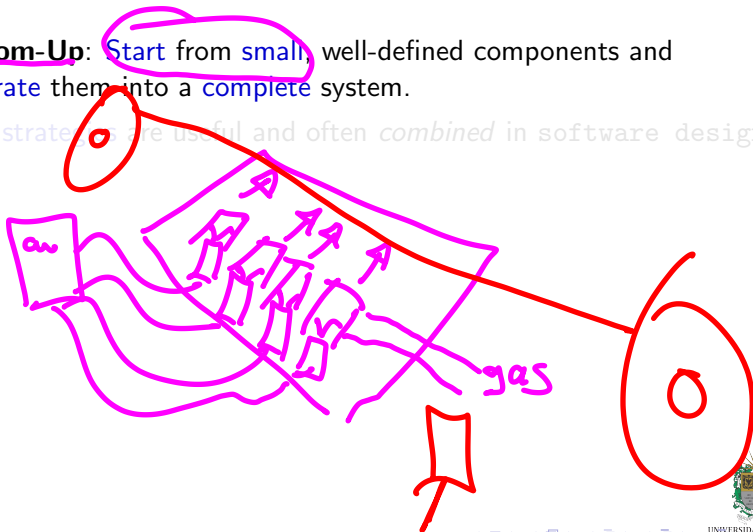
Strategies to Solve Problems

- **Top-Down:** Start from the **big** picture and **break** it down into smaller parts.
- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.
- Both strategies are useful and often **combined** in software design.



Strategies to Solve Problems

- **Top-Down:** Start from the **big** picture and **break** it down into smaller parts.
- **Bottom-Up:** Start from **small**, well-defined components and **integrate** them into a **complete** system.
- Both strategies are useful and often *combined* in software design.



Strategies to Solve Problems

- **Top-Down:** Start from the big picture and break it down into smaller parts.
- **Bottom-Up:** Start from small, well-defined components and integrate them into a complete system.
- Both strategies are useful and often *combined* in software design.



Outline

- 1 Creating Models in Design
- 2 Evolution of Programming Languages
- 3 Four Design Principles**
- 4 SOLID Principles



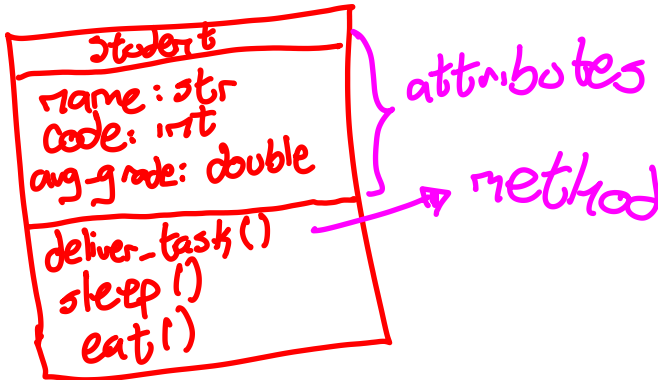
Object-Oriented Design and Contracts

- **Object-Oriented Design (OOD)** organizes software as a **collection of objects**.
- **Contracts**: Define responsibilities and expectations between objects.
- **Contracts** help ensure correctness and robustness.



Object-Oriented Design and Contracts

- **Object-Oriented Design (OOD)** organizes software as a **collection of objects**.
- **Contracts:** Define responsibilities and **expectations** between objects.
- **Contracts** help ensure **correctness** and **robustness**.



UML Diagrams

- **UML** (*Unified Modeling Language*) is a standard way to visualize system design.

- Common diagrams: Class diagrams, Sequence diagrams, and Use case diagrams

- UML helps communicate design as clearly

Handwritten notes in blue ink:

- 34 → C/ass Diagrams
- 46 → Sequence diagrams
- Delay rest " " → Use case diagrams
- support
- standard
- Activities
- use cases



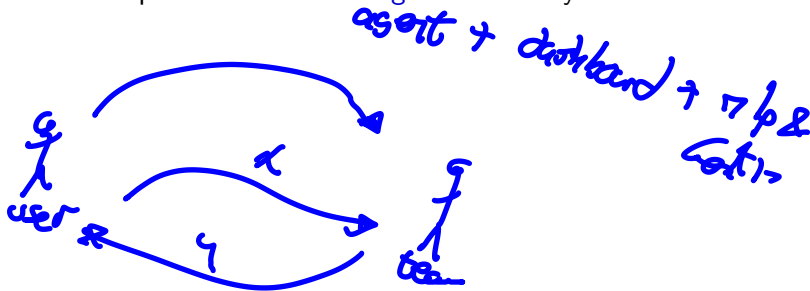
UML Diagrams

- **UML** (*Unified Modeling Language*) is a standard way to **visualize** system design.]
- **Common diagrams:** Class diagrams, Sequence diagrams, and Use case diagrams. } **oop** **calls**
- **Requirements**
- UML helps communicate design ideas clearly.



UML Diagrams

- **UML** (*Unified Modeling Language*) is a standard way to **visualize** system design.
- **Common diagrams**: Class diagrams, Sequence diagrams, and Use case diagrams.
- **UML** helps **communicate** design ideas clearly.



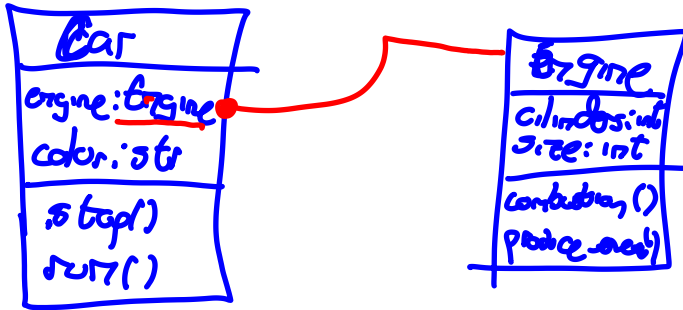
Class Diagrams

- Class diagrams show the structure of the system.
- They display **classes**, their attributes, methods, and relationships.
- Useful for both conceptual and technical design.



Class Diagrams

- **Class diagrams** show the **structure of the system**.
- They display **classes**, their **attributes**, **methods**, and **relationships**.
- **Useful** for both **conceptual** and **technical design**.



Abstraction

- **Abstraction** means focusing on the essential features of an object.
- **Rule of Least Astonishment:** Design so users are not surprised by behavior.
- Consider context, basic attributes, and basic behaviors when *designing abstractions*.

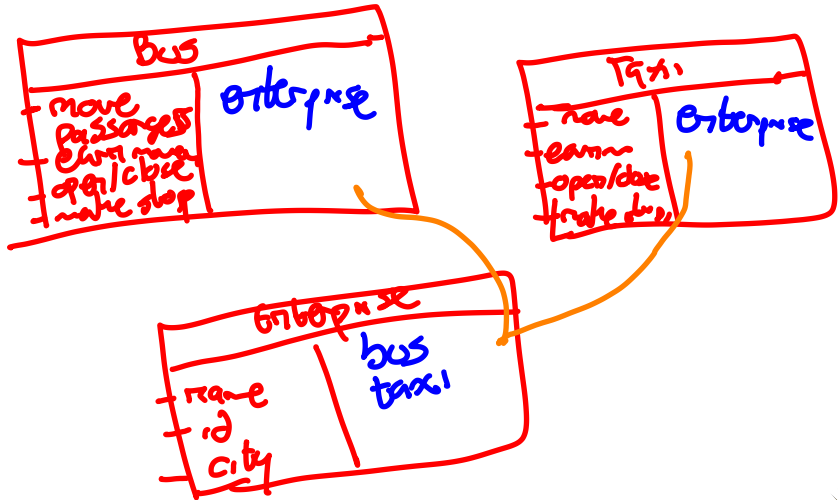


Abstraction

- **Abstraction** means focusing on the **essential features** of an object.
- **Rule of Least Astonishment:** Design so **users** are **not surprised** by behavior.
- Consider **context**, **basic attributes**, and **basic behaviors** when *designing abstractions*.



Abstraction & CRC Cards



Encapsulation

- **Encapsulation** bundles attributes and methods together.



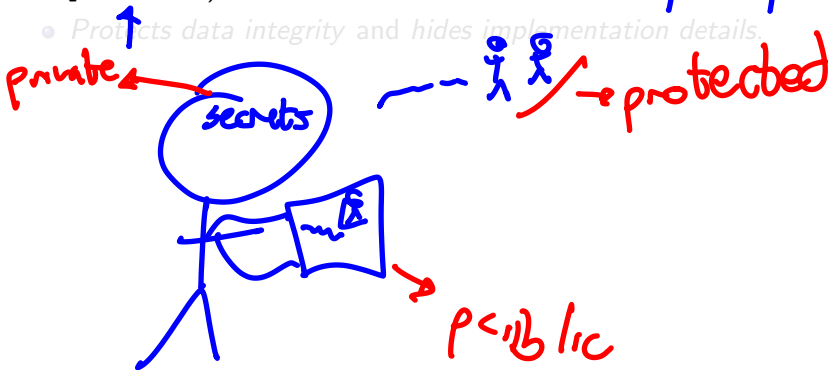
hide

information



Encapsulation

- **Encapsulation** bundles **attributes** and **methods** together.
- Expose only what is necessary (access levels: public, private, protected).
 - ↑
 - ↑
- *Protects data integrity and hides implementation details.*



Encapsulation

- **Encapsulation** bundles **attributes** and **methods** together.
- Expose **only what is necessary** (access levels: public, private, protected).
- *Protects data integrity and hides implementation details.*

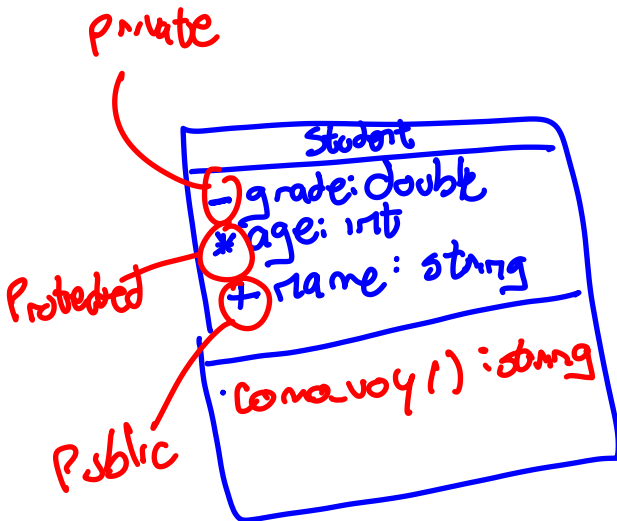
```

public class Student {
    private double grade;
    protected int age;

    public String comp-avg() {
        if (grade >= 3.0) {
            return "B+ or better";
        } else if (grade >= 2.0) {
            return "C or better";
        } else {
            return "Below C";
        }
    }
}
  
```



Encapsulation & UML



Black Box Thinking

- **Objects communicate** through **well-defined interfaces**.
- **Rule of Least Knowledge**: Objects should **know** as **little** as **possible** about one another.
- **Black box**: Focus on what an object does, not how it does it.

Demo time!



Black Box Thinking

- **Objects communicate** through **well-defined interfaces**.
- **Rule of Least Knowledge**: Objects should **know** as **little as possible** about one another.
- **Black box**: Focus on what an object does, **not how it does it**.

Demo time!



Data Integrity: Getters and Setters

- **Getters** and **Setters** are methods to access and modify object attributes.
- They help maintain **data integrity** by controlling how attributes are accessed and modified.
- Use them to enforce **validation rules** and **business logic**.

Demo time!



Data Integrity: Getters and Setters

- **Getters** and **Setters** are methods to access and modify object attributes.
- They help maintain **data integrity** by controlling how attributes are accessed and modified.
- Use them to enforce **validation rules** and **business logic**.

Demo time!



Decomposition

- **Decomposition:** Divide and conquer by breaking the system into smaller parts.
- Separation of Concerns: Each part should have a clear responsibility.
- Consider **object** lifetime, associations, aggregation, and composition.



Decomposition

- **Decomposition:** Divide and conquer by breaking the system into smaller parts.
- **Separation of Concerns:** Each part should have a clear responsibility.
- Consider object lifetime, associations, aggregation, and composition.



Decomposition

- **Decomposition:** Divide and conquer by breaking the system into smaller parts.
- **Separation of Concerns:** Each part should have a clear responsibility.
- Consider **object** lifetime, associations, aggregation, and composition.



Decomposition Example: Kitchen in a House



Association

A **relationship** between two classes where **one class** uses or **interacts** with **another class**.



Aggregation

An **aggregation** is a **whole-part relationship** where one class is a **part** of another class, but **can exist independently**.

Demo time!



Aggregation

An **aggregation** is a **whole-part relationship** where one class is a **part** of another class, but **can exist independently**.

Demo time!



Composition

Composition is a **stronger whole-part relationship** where one class is a **part** of another class and **cannot exist independently**.

Demo time!



Composition

Composition is a **stronger whole-part relationship** where one class is a **part** of another class and **cannot exist independently**.

Demo time!



Generalization

- **Generalization** eliminates redundancy by extracting **common features**.
- **D.R.Y. Principle:** Don't Repeat Yourself.
- Behaviors can be generalized using **inheritance**, **interface inheritance**, and **abstract classes**.
- **Polymorphism:** Objects can be treated as **instances** of their parent class.
- **Types of inheritance:** single, multiple, interface-based.



Generalization

- **Generalization** eliminates redundancy by extracting **common features**.
- **D.R.Y. Principle:** Don't Repeat Yourself.
- Behaviors can be generalized using **inheritance**, **interface inheritance**, and **abstract classes**.
- **Polymorphism:** Objects can be treated as **instances** of their parent class.
- **Types of inheritance:** single, multiple, interface-based.



Generalization

- **Generalization** eliminates redundancy by extracting **common features**.
- **D.R.Y. Principle**: Don't Repeat Yourself.
- Behaviors can be generalized using **inheritance**, **interface inheritance**, and **abstract classes**.
- **Polymorphism**: Objects can be treated as **instances** of their parent class.
- **Types of inheritance**: single, multiple, interface-based.



Generalization

- **Generalization** eliminates redundancy by extracting **common features**.
- **D.R.Y. Principle**: Don't Repeat Yourself.
- Behaviors can be generalized using **inheritance**, **interface inheritance**, and **abstract classes**.
- **Polymorphism**: Objects can be treated as **instances** of their parent class.
- **Types of inheritance**: single, multiple, interface-based.



Inheritance

- **Inheritance** allows a class to inherit properties and methods from another class.
- **Base class**: The class being inherited from.
- **Derived class**: The class that inherits from the base class.
- **Benefits**: Code reusability, easier maintenance, and **polymorphism**.
- **Drawbacks**: Complexity, tight coupling, and potential **fragility**.

Demo time!



Inheritance

- **Inheritance** allows a class to inherit properties and methods from another class.
- **Base class:** The class being inherited from.
- **Derived class:** The class that inherits from the base class.
- **Benefits:** Code reusability, easier maintenance, and **polymorphism**.
- **Drawbacks:** Complexity, tight coupling, and potential **fragility**.

Demo time!



Inheritance

- **Inheritance** allows a class to inherit properties and methods from another class.
- **Base class:** The class being inherited from.
- **Derived class:** The class that inherits from the base class.
- **Benefits:** Code reusability, easier maintenance, and [polymorphism](#).
- **Drawbacks:** Complexity, tight coupling, and potential [fragility](#).

Demo time!



Inheritance & UML



Interface Inheritance

- **Interface inheritance** allows a class to implement an interface *without inheriting its implementation*.
- Interfaces define a *contract* that classes must adhere to.
- **Benefits:** Flexibility, code reusability, and easier maintenance.
- **Drawbacks:** Complexity and potential performance issues.

Demo time!



Interface Inheritance

- **Interface inheritance** allows a class to implement an interface *without inheriting its implementation*.
- **Interfaces** define a *contract* that classes must adhere to.
- **Benefits:** Flexibility, code reusability, and easier maintenance.
- **Drawbacks:** Complexity and potential performance issues.

Demo time!



Interface Inheritance

- **Interface inheritance** allows a class to implement an interface *without inheriting its implementation*.
- **Interfaces** define a *contract* that classes must adhere to.
- **Benefits:** Flexibility, code reusability, and easier maintenance.
- **Drawbacks:** *Complexity* and potential *performance* issues.

Demo time!



Polymorphism by Inheritance

- **Polymorphism** allows objects of different classes to be treated as objects of a common superclass.
- **Benefits:** Code reusability, flexibility, and easier maintenance.
- **Drawbacks:** Complexity and potential performance issues.

Demo time!



Polymorphism by Inheritance

- **Polymorphism** allows objects of different classes to be treated as **objects of a common superclass**.
- **Benefits:** Code reusability, flexibility, and easier maintenance.
- **Drawbacks:** **Complexity** and potential **performance** issues.

Demo time!



Polymorphism by Overloading

- **Polymorphism by overloading** allows **multiple methods** with the **same name** but different parameters.
- **Benefits:** Improves readability and reduces complexity.
- **Drawbacks:** Can lead to **confusion** if not used carefully.

Demo time!



Polymorphism by Overloading

- **Polymorphism by overloading** allows **multiple methods** with the **same name** but different parameters.
- **Benefits:** **Improves readability** and **reduces complexity**.
- **Drawbacks:** Can lead to **confusion** if not used carefully.

Demo time!



Outline

- 1 Creating Models in Design
- 2 Evolution of Programming Languages
- 3 Four Design Principles
- 4 SOLID Principles**



SOLID Principles

- **Single Responsibility Principle (SRP):** A class should have only one reason to change.
- **Open/Closed Principle (OCP):** A class should be open for extension, but closed for modification.
- **Liskov Substitution Principle (LSP):** Objects in a program should be replaceable with instances of their subtypes without altering the correctness of that program.
- **Interface Segregation Principle (ISP):** A client should never be forced to implement an interface that it doesn't use or clients shouldn't be forced to depend on methods they do not use.
- **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules. Both should depend on abstractions. Abstractions should not depend on details. Details should depend on abstractions.



SOLID Principles

- **Single Responsibility Principle (SRP):** A class should have only **one reason to change**.
- **Open/Closed Principle (OCP):** A class should be **open** for extension, but **closed** for modification.
- **Liskov Substitution Principle (LSP):** Objects in a program should be replaceable with instances of their **subtypes** without altering the correctness of that program.
- **Interface Segregation Principle (ISP):** A client **should never be forced** to implement an interface that it doesn't use or clients **shouldn't be forced** to depend on methods they do not use.
- **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules. Both should **depend on** **abstractions**. Abstractions should not depend on details. Details should **depend on** **abstractions**.



SOLID Principles

- **Single Responsibility Principle (SRP):** A class should have only **one reason to change**.
- **Open/Closed Principle (OCP):** A class should be **open** for extension, but **closed** for modification.
- **Liskov Substitution Principle (LSP):** Objects in a program should be replaceable with instances of their **subtypes** without altering the correctness of that program.
- **Interface Segregation Principle (ISP):** A client **should never be forced** to implement an interface that it doesn't use or clients **shouldn't be forced** to depend on methods they do not use.
- **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules. Both should **depend on abstractions**. Abstractions should not depend on details. **Details should depend on abstractions**.



SOLID Principles

- **Single Responsibility Principle (SRP):** A class should have only **one reason to change**.
- **Open/Closed Principle (OCP):** A class should be **open** for extension, but **closed** for modification.
- **Liskov Substitution Principle (LSP):** Objects in a program should be replaceable with instances of their **subtypes** without altering the correctness of that program.
- **Interface Segregation Principle (ISP):** A client **should never be forced** to implement an interface that it doesn't use or clients **shouldn't be forced** to depend on methods they do not use.
- **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules. Both should **depend on abstractions**. Abstractions should not depend on details. **Details should depend on abstractions**.



SOLID Principles

- **Single Responsibility Principle (SRP):** A class should have only **one reason to change**.
- **Open/Closed Principle (OCP):** A class should be **open** for extension, but **closed** for modification.
- **Liskov Substitution Principle (LSP):** Objects in a program should be replaceable with instances of their **subtypes** without altering the correctness of that program.
- **Interface Segregation Principle (ISP):** A client **should never be forced** to implement an interface that it doesn't use or clients **shouldn't be forced** to depend on methods they do not use.
- **Dependency Inversion Principle (DIP):** High-level modules should not depend on low-level modules. Both should **depend on abstractions**. Abstractions should not depend on details. **Details should depend on abstractions**.



Good Practices

- **Composition over Inheritance:** Inheritance should be used **only when there is a clear relationship** between the base class and the derived class. In other cases, **composition** should be used. Inheritance is a powerful tool, but it is not always the best tool for the job. Inheritance is a way to achieve polymorphism, but it is **not the only way to achieve polymorphism**.
- **Code to Interfaces, not Implementations:** This principle is about designing your classes so that they **depend on interfaces** rather than concrete classes.

Demo time!



Good Practices

- **Composition over Inheritance:** Inheritance should be used **only when there is a clear relationship** between the base class and the derived class. In other cases, **composition** should be used. Inheritance is a powerful tool, but it is not always the best tool for the job. Inheritance is a way to achieve polymorphism, but it is **not the only way to achieve polymorphism**.
- **Code to Interfaces, not Implementations:** This principle is about designing your classes so that they **depend on interfaces** rather than concrete classes.

Demo time!



Outline

- 1 Creating Models in Design
- 2 Evolution of Programming Languages
- 3 Four Design Principles
- 4 SOLID Principles



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/object-oriented-programming>

